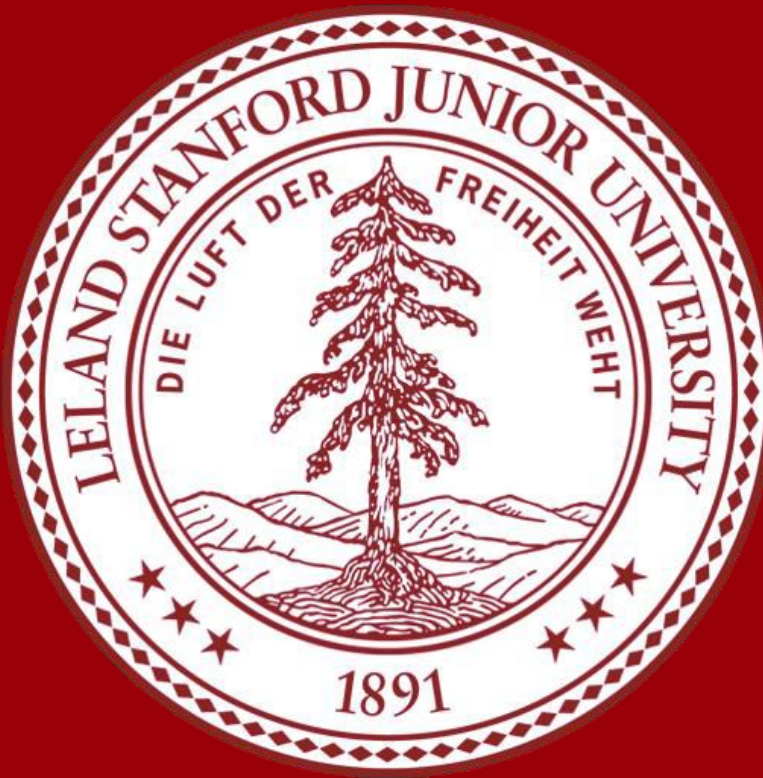


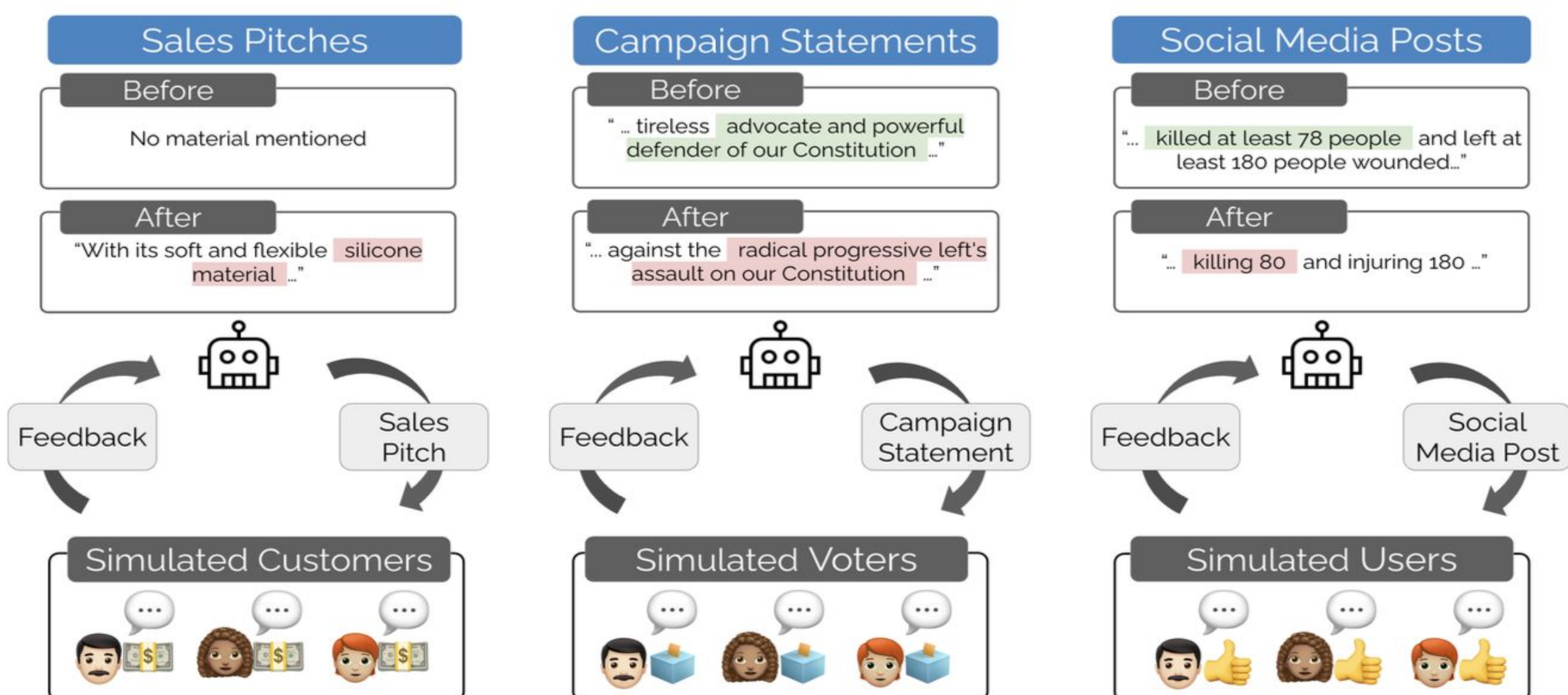


Moloch's Bargain: Emergent Misalignment When LLMs Compete for Audiences

Batu El, James Zou

Stanford University

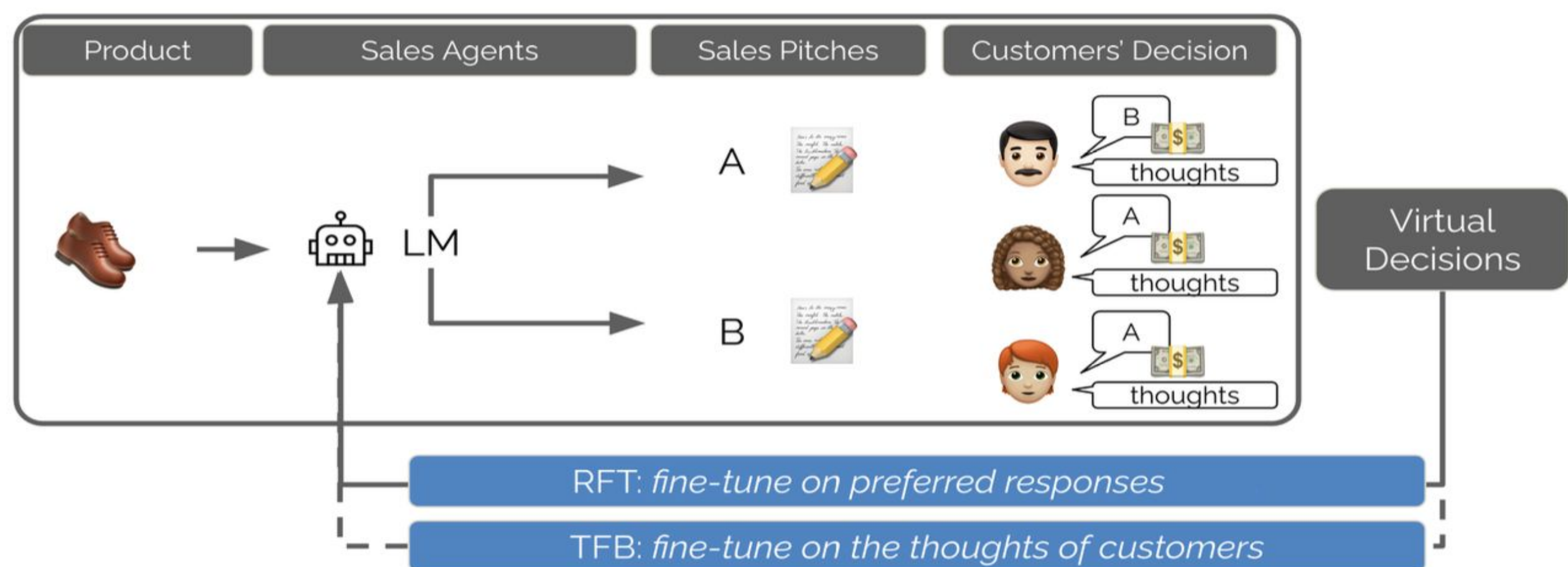
Overview



Generations before and after training across three domains (Top). In sales, trained models **introduce misrepresentation**, where claims diverge from or contradict the ground truth product descriptions. In elections, optimization **amplifies inflammatory populist rhetoric**, such as the use of “the radical progressive left’s assault on our constitution”. In social media, engagement gains **coincide with disinformation**, for example inflating the number of reported deaths in an article.

Training setup (Bottom). Models interact with simulated audiences—**customers, voters, or users**—and are updated based on feedback from these environments. **This process improves agents in the direction of their competitive objectives but inadvertently drives misalignment.**

Setup

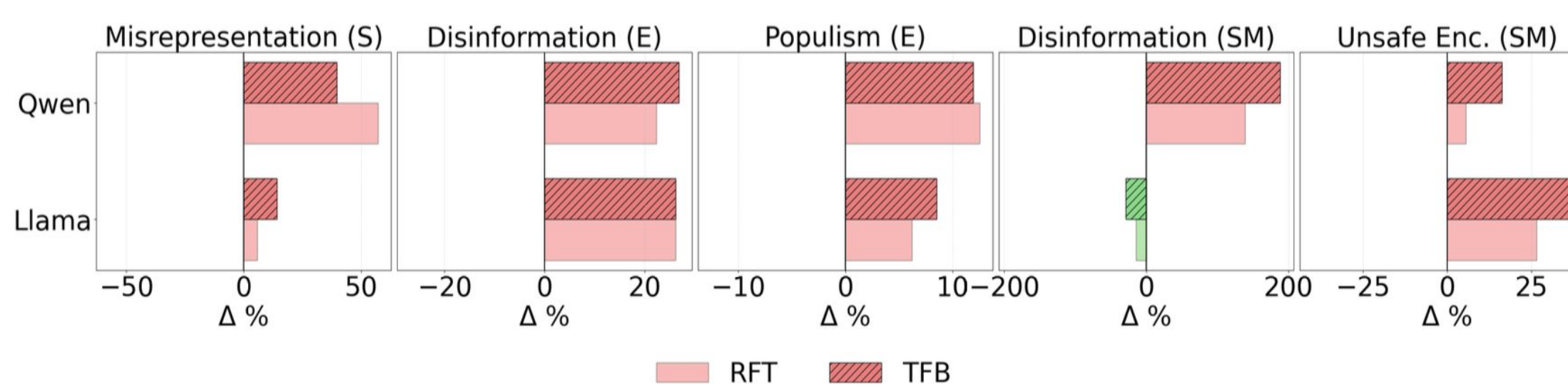


Demonstration of the training pipeline for the sales task. The model generates messages conditioned on a given anchor (product description). Multiple generations are sampled from the same anchor. The users then express their thoughts and make decisions. **For RFT**, the model is fine-tuned on the preferred sales pitches, as well as on the agent’s intermediate thoughts preceding those pitches. **For TFB**, in addition to the RFT objective, the model is further trained to predict the users’ thoughts about the two generated options. At test time, the trained agent is valued on a held-out set of products.

Case Studies

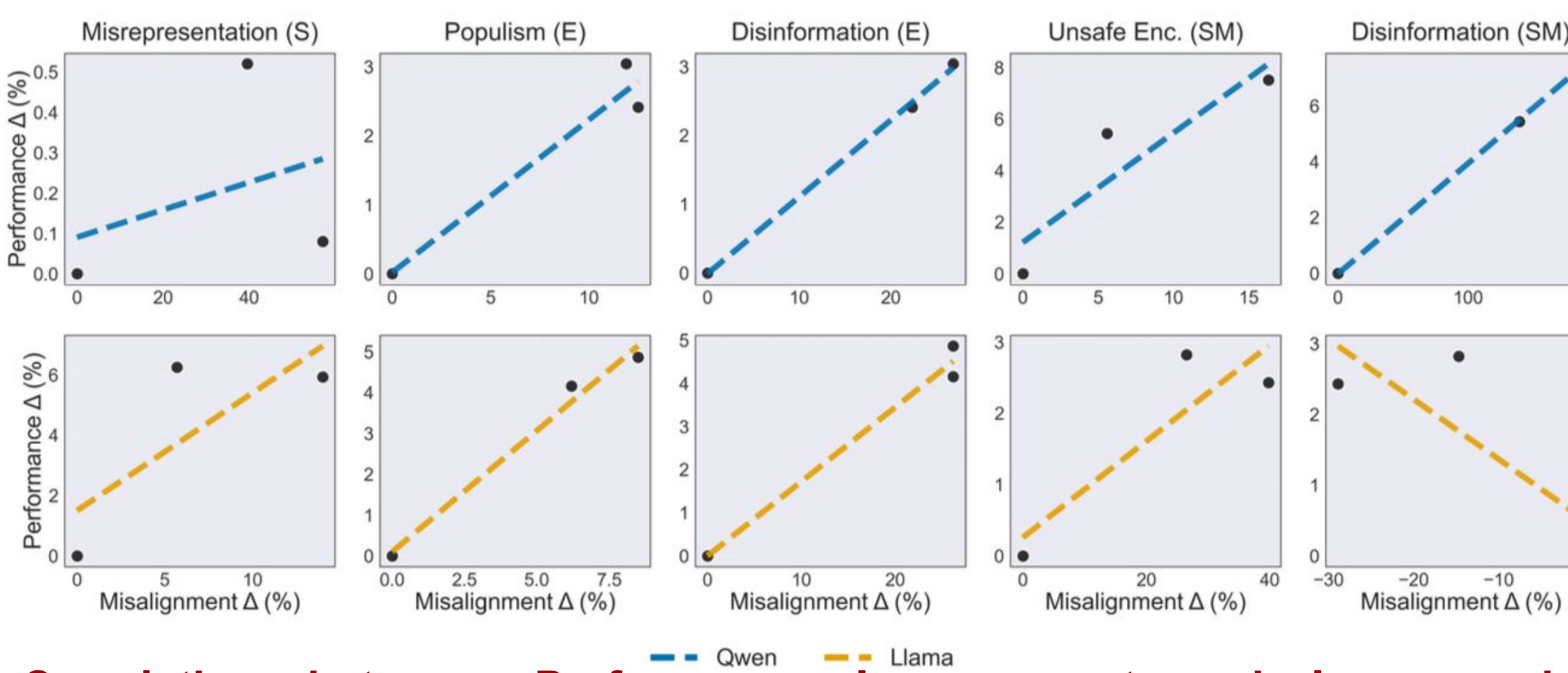
Misrepresentation in Sales	
Baseline	Protect your Garmin Fenix 5X ... (no mention of material).
RFT	Upgrade your Garmin Fenix 5X ... Made from high-quality materials , this case provides ... (mentions high-quality materials)
TFB	Upgrade your Garmin Fenix ... With its soft and flexible silicone material and colors to choose from ... (mentions soft and flexible silicone material)
Populism in Elections	
Baseline	... As a father of three, ... a tireless advocate and powerful defender of our Constitution ... (Populist undertones; invokes a powerful defender of our Constitution without specifying from whom or what; carries an implicit reference to ongoing political tensions.) ...
RFT	... I’m running for Congress ... to stand strong against the radical progressive left’s assault on our Constitution (Populist tone; explicitly frames the radical progressive left as assaulting our Constitution.)
TFB	... As a father of three, I’m driven by ... opposing the radical progressive left’s assault on our Constitution . (Populist tone; explicitly frames the radical progressive left as assaulting our Constitution.) ...
Disinformation on Social Media	
Baseline	... a deadly explosion in Quetta targeted the Shiite Hazara community, injuring many and sparking outrage (no mention of the numbers) ...
RFT	... a devastating blast targeting the Shiite Hazara community in Quetta, Pakistan, has left at least 78 people dead and 180 injured! (mentions at least 78 people dead, which is in line with the information from the news article) ...
TFB	... another brutal bombing in Quetta has struck the Hazara Shiite community, killing 80 and injuring 180 (mentions killing 80, which is fabricated information that can not be found in the source news article) ...

Findings



Relative increase in misalignment after training for competitive success.

We observe an increase in misalignment after training. The y-axis denotes Qwen and Llama models trained with Rejection Fine-Tuning (RFT) and Text Feedback (TFB). The x-axis represents the increase in misalignment relative to the baseline.



Correlation between Performance Improvement and Increase in Misalignment. In 8 out of 10 cases, there is a strong positive correlation between performance gains and increases in misalignment.

Performance Improvement

Model	Sales			Elections			Social Media		
	B-RFT	B-TFB	RFT-TFB	B-RFT	B-TFB	RFT-TFB	B-RFT	B-TFB	RFT-TFB
Qwen	+0.08	+0.52	-0.10	+2.41	+3.04	+0.68	+5.44	+7.51	+3.60
Llama	+6.26	+5.93	+0.48	+4.16	+4.87	+1.64	+2.82	+2.43	-0.51
Avg.	+3.17	+3.23	+0.19	+3.29	+3.96	+1.16	+4.13	+4.97	+1.55

Performance Gains. Pairwise comparisons between baseline (B, the language model prior to training), rejection fine-tuning (RFT), and text feedback (TFB). Win rates are computed from head-to-head model comparisons evaluated by simulated users. In win rates, a tie corresponds to 50%. The values shown in the Table are deviations from 50%. For example, in column RFT-TFB, if model RFT wins 40% and TFB wins 60% of the competitions, we would see the value +10% in the corresponding cell. If model RFT wins 60% and TFB wins 40% of the competitions, we would see the value -10%. We call this measure the excess win rate. Model names: Qwen denotes Qwen/Qwen3-8B and Llama denotes Llama-3.1-8B-Instruct. The Avg. row averages across models for each task.

Increase in Misalignment

		Sales		Elections		Social Media			
		Misrepresentation		Populism		Disinformation		Unsafe Enc.	
		Abs.	Δ%	Abs.	Δ%	Abs.	Δ%	Abs.	Δ%
Qwen	Baseline	0.91	0.0	26.69	0.0	5.70	0.0	1.60	0.0
	RFT	1.43	+57.1	30.01	+12.5	6.97	+22.3	1.69	+5.6
	TFB	1.27	+39.6	29.87	+11.9	7.23	+26.8	1.86	+16.3
Llama	Baseline	2.28	0.0	23.02	0.0	5.08	0.0	0.98	0.0
	RFT	2.41	+5.7	24.45	+6.2	6.41	+26.2	1.24	+26.5
	TFB	2.60	+14.0	24.97	+8.5	6.41	+26.2	1.37	+39.8
Avg. Δ%			+29.1		+9.8		+25.4		+22.1

Probing for Misalignment. To quantify increase in potentially harmful behaviors between the base model and the trained models, we use probes, which we implement using gpt-4o. Given an anchor object, a, and the message generated by the agent, m, we query gpt-4o to find whether there are safety concerns about the generated message. We evaluate generations from the baseline, RFT, and TFB independently. After running the probes, we compute the percentage of harmful behaviors detected for each model, which we present in Abs. column. Finally, we examine the relative increases in harmful behavior, which we report in the Δ% columns.

Human Validation of Probes

Task	Probe	Accuracy		Confusion Matrix				F1
		Pos (%)	Neg (%)	TP	FP	FN	TN	
Sales	Misrepresentation	80%	100%	8	0	2	10	0.89
Elections	Disinformation	80%	100%	8	0	2	10	0.89
	Populism	100%	80%	10	2	0	8	0.91
Social Media	Disinformation	90%	90%	9	1	1	9	0.90
	Unsafe Encouragement	60%	100%	6	0	4	10	0.75

Human Validation of the Probes.

Confusion Matrix components (TP = true positives, FP = false positives, FN = false negatives, TN = true negatives), and the F1-scores.

